

Azure Data Factory

Azure Data Factory (ADF) Complete Documentation

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1. Introduction to Azure Data Factory

Azure Data Factory (ADF) is Microsoft's cloud-based data integration service used for:

- Data movement
- Data transformation
- Data orchestration
- ETL and ELT processing
- Scheduling and automation
- Building data pipelines

ADF helps organizations move data from multiple sources into data warehouses, data lakes, databases, APIs, and analytics platforms.

ADF is mainly used by:

- Data Engineers
- BI Engineers
- Cloud Engineers
- Analytics Teams
- Data Architects

2. Why Use ADF

Advantages of ADF

- Fully cloud-based
- Scalable architecture
- Supports 100+ connectors
- Low-code/no-code environment
- Easy scheduling
- Supports ETL and ELT
- Cost-effective
- Enterprise-grade security
- Supports hybrid integration
- Easy monitoring

Real-Time Uses

- Daily sales data loading
- Banking transaction processing
- API ingestion
- Data warehouse loading
- Data lake ingestion
- CRM data synchronization
- ERP integration
- Log processing

3. Core Concepts of ADF

Main Building Blocks

Component	Description
Pipeline	Group of activities
Activity 	Single task inside pipeline
Dataset	Data structure reference
Linked Service	Connection string/configuration
Trigger	Executes pipeline automatically
Integration Runtime	Compute infrastructure
Data Flow	Visual transformation engine

4. ADF Architecture

ADF Architecture Flow

Source System → Linked Service → Dataset → Pipeline → Activity → Destination

Layers

Source Layer

- SQL Server
- Oracle
- SAP
- REST API
- Blob Storage
- AWS S3
- FTP
- Excel
- CSV

Processing Layer

- Copy Activity
- Data Flow
- Databricks
- Stored Procedures

Destination Layer

- Azure SQL
- Synapse
- Data Lake
- Snowflake
- Cosmos DB

5. Azure Data Factory Components

5.1 Pipelines

Pipeline is a logical grouping of activities.

Example:

- Extract customer data
- Transform data
- Load into SQL database
- Send notification email

5.2 Activities

Activities perform actual work.

Types of Activities

Activity Type	Purpose
Copy Activity	Move data
Data Flow	Transform data
Lookup	Read small data
ForEach	Loop processing
If Condition	Conditional execution
Stored Procedure	Execute SQL procedure
Web Activity	Call APIs
Execute Pipeline	Call another pipeline

5.3 Datasets

Datasets represent the structure of data.

Examples:

- CSV file
- SQL table
- JSON file
- Parquet file

5.4 Linked Services

Linked services define connection information.

Examples:

- SQL connection
- Blob storage connection
- REST API connection

5.5 Triggers

Triggers automate pipeline execution.

Types:

- Schedule Trigger
- Tumbling Window Trigger
- Event Trigger

6. Azure Portal Basics

Important Azure Services

Service	Purpose
Resource Group	Container for resources
Storage Account	Store files/data
Azure SQL Database	Database service
Key Vault	Secrets management
Virtual Network	Networking
Monitor	Logging and monitoring

7. Creating Azure Free Account

Steps

1. Go to Azure official website
2. Create Microsoft account
3. Verify mobile number
4. Add card details
5. Activate free subscription

Free Services

- Free credits
- Limited free storage
- Free databases
- Limited compute

8. Creating Azure Data Factory Instance

Steps

1. Login to Azure Portal
 2. Search for Azure Data Factory
 3. Click Create
 4. Select subscription
 5. Select resource group
 6. Enter factory name
 7. Select region
 8. Click Review + Create
 9. Click Create
-

9. Understanding Integration Runtime

Integration Runtime (IR) is the compute infrastructure used by ADF.

Types of IR

Type	Purpose
Azure IR	Cloud data movement
Self-Hosted IR	On-premise connectivity
Azure SSIS IR	Execute SSIS packages

Self-Hosted IR Usage

Used when:

- Data is inside local servers
- Firewall restrictions exist
- Private network access required

10. Linked Services

Linked service acts like a connection manager.

Common Linked Services

- Azure Blob Storage
- Azure SQL Database
- SQL Server
- Oracle
- Snowflake
- REST API
- Amazon S3
- FTP/SFTP

Steps to Create Linked Service

1. Open Manage Hub
 2. Click Linked Services
 3. Click New
 4. Select connector
 5. Enter credentials
 6. Test connection
 7. Create
-

11. Datasets

Dataset defines data structure.

Dataset Examples

Dataset Type	Example
Delimited Text	CSV file
JSON	API response
Parquet	Optimized analytics file
SQL Table	Employee table

Parameters in Dataset

Datasets can use dynamic values.

Example:

- Dynamic file names
- Dynamic table names
- Dynamic folder paths

12. Pipelines

Pipeline is the main orchestration component.

Pipeline Features

- Sequential execution
- Parallel execution
- Conditional execution
- Dynamic execution
- Scheduling

Real-Time Example

1. Read CSV file
 2. Validate data
 3. Load to staging table
 4. Execute stored procedure
 5. Send success email
-

13. Activities in ADF

13.1 Copy Activity

Used to move data.

Supported Sources

- SQL
- Oracle
- Blob
- S3
- FTP
- APIs

Supported Destinations

- Synapse
- SQL Database
- Data Lake
- Snowflake

13.2 Lookup Activity

Used to fetch small amounts of data.

13.3 Get Metadata Activity

Used to retrieve:

- File names
- File size
- Last modified date



13.4 Delete Activity

Deletes files/folders.

13.5 Validation Activity

Validates file existence.

14. Copy Data Activity

Copy activity is the most used activity.

Working Flow

Source → Staging → Destination

Features

- Parallel copy
- Schema mapping
- Fault tolerance
- Compression support
- Incremental copy

Copy Activity Settings

Setting	Purpose
Source	Input data
Sink	Destination
Mapping	Column mapping
Fault tolerance	Skip errors
Staging	Temporary storage

15. Data Flow

Data Flow is used for visual data transformation.

Types

Type	Purpose
Mapping Data Flow	Transformation
Wrangling Data Flow	Power Query-based

Common Transformations

- Filter
- Derived Column
- Join
- Aggregate
- Sort
- Union
- Conditional Split
- Lookup
- Window

Data Flow Architecture

Source → Transformation → Sink

16. Control Flow Activities

Important Control Activities

Activity	Purpose
If Condition	Conditional logic
Switch	Multiple conditions
Until	Loop until condition met
ForEach	Iterate collections
Wait	Delay execution
Fail	Fail pipeline intentionally

17. Parameters and Variables

Parameters

Used to pass values dynamically.

Example

- File name
- Table name
- Date value

Variables

Used to store values temporarily.

Variable Types

- String
 - Boolean
 - Array
 - Integer
-

18. Expressions and Dynamic Content

ADF uses dynamic expressions.

Common Functions

Function	Purpose
concat()	Combine strings
utcNow()	Current UTC time
pipeline()	Pipeline metadata
variables()	Access variable
activity()	Activity output
substring()	Extract text

Example

```
@concat('File_',utcNow(),'.csv')
```

19. Triggers

Triggers automate execution.

19.1 Schedule Trigger

Runs pipeline at fixed intervals.

Example:

- Daily at 9 PM
- Every 15 minutes



19.2 Tumbling Window Trigger

Processes data in chunks/windows.

Used in incremental loading.

19.3 Event Trigger

Executes when:

- File arrives
- Blob created
- Blob deleted

20. Monitoring and Debugging

Monitoring Options

- Pipeline runs
- Activity runs
- Trigger runs
- Error logs

Debug Mode

Used for testing pipelines.

Monitoring Metrics

- Duration
- Throughput
- Queue time
- Rows copied

21. Error Handling

Common Techniques

Try-Catch Pattern

Use:

- Execute pipeline
- Failure path
- Success path



Logging Errors

Store errors into:

- SQL table
- Blob storage
- Log analytics

Retry Mechanism

ADF supports retries.

Example:

- Retry count = 3
- Retry interval = 30 sec

22. Incremental Load

Incremental load copies only new or changed data.

Advantages

- Faster execution
- Lower cost
- Reduced load on systems

Common Approaches

Method	Description
Watermark	Based on date/time
CDC	Change Data Capture
Timestamp	Last modified value
Identity Column	Max ID tracking

Incremental Load Flow

1. Get max timestamp
2. Fetch new data
3. Load into target
4. Update watermark table

23. Full Load



Full load copies complete data every time.

Advantages

- Simpler logic
- Easy implementation

Disadvantages

- Higher execution time
- Higher cost
- Duplicate risk

24. ETL vs ELT in ADF

Feature	ETL	ELT
Transformation	Before loading	After loading
Performance	Slower	Faster
Compute	External	Target system
Example	SSIS	Synapse

ETL Flow

Extract → Transform → Load

ELT Flow

Extract → Load → Transform

25. Connecting Different Data Sources

Supported Sources

- SQL Server
- Oracle
- MySQL
- PostgreSQL
- SAP
- Blob Storage
- S3
- Google Cloud Storage
- REST APIs
- Salesforce
- Service



26. SQL Integration with ADF

Common SQL Operations

- Execute stored procedures
- Read SQL tables
- Insert/update data
- Merge operations

Stored Procedure Activity

Used to execute SQL procedures.

Example Use Cases

- Audit logging
- Data validation
- Merge operations
- Data cleansing

27. File Handling in ADF

Supported File Formats

Format	Usage
CSV	Flat file
JSON	Semi-structured
XML	Hierarchical
Parquet	Analytics
Avro	Big data
Excel	Reporting

Compression Support

- ZIP
- GZIP
- BZIP2

28. REST API Integration

ADF can connect to APIs.

Steps

1. Create REST linked service
2. Configure authentication
3. Create dataset
4. Use copy activity

Authentication Types

- Anonymous
- Basic
- OAuth2
- Managed Identity

29. Scheduling Pipelines

Scheduling Methods

Method	Example
Hourly	Every 1 hour
Daily	Daily at 10 PM
Weekly	Every Sunday
Monthly	1st day of month

30. CI/CD in ADF

CI/CD means Continuous Integration and Continuous Deployment.

Benefits

- Automated deployment
- Version control
- Faster release cycles
- Reduced manual effort

Tools Used

- Azure DevOps
- GitHub
- ARM Templates



31. Git Integration

ADF supports Git integration.

Supported Repositories

- Azure DevOps Git
- GitHub

Benefits

- Version control
- Collaboration
- Rollback support

- Branching

32. ARM Templates

ARM templates are used for deployment automation.

Uses

- Infrastructure deployment
- Environment migration
- Backup configuration

33. Security in ADF

Security Features

- Encryption
- RBAC
- Managed Identity
- Private endpoints
- Key Vault integration

34. Managed Identity

Managed identity removes need for storing passwords.

Benefits



- Secure authentication
- Passwordless access
- Easy integration

35. Role-Based Access Control (RBAC)

RBAC controls user permissions.

Common Roles

Role	Access
Owner	Full access

Role	Access
Contributor	Modify resources
Reader	Read-only
Data Factory Contributor	ADF access

36. Azure Key Vault Integration

Key Vault stores secrets securely.

Stored Items

- Passwords
- API keys
- Connection strings
- Certificates

Advantages

- Improved security
- Centralized secrets management

37. Performance Optimization

Optimization Techniques

- Parallel copy
- Partitioning
- Compression
- Incremental loading
- Proper IR selection
- Optimize data flow clusters

Performance Metrics

- DIU usage
- Throughput
- Queue duration

38. Logging and Monitoring

Monitoring Tools

- Azure Monitor
- Log Analytics
- Diagnostic logs
- Alerts

Alert Examples

- Pipeline failure
- High execution time
- Integration runtime offline

39. Data Lake Integration

ADF integrates with Azure Data Lake.

Advantages

- Scalable storage
- Cheap storage
- Big data processing

Common Architecture

Source → ADF → Data Lake → Synapse/Databricks

40. Synapse Integration

ADF works closely with Azure Synapse.

Use Cases



- Data warehouse loading
- ELT processing
- Analytics pipelines

41. Databricks Integration

ADF can execute Databricks notebooks.

Common Use Cases

- PySpark processing
- Machine learning
- Big data transformation

42. Snowflake Integration

ADF supports Snowflake connectivity.

Operations

- Data loading
- Data extraction
- Stored procedure execution

43. SAP Integration

ADF supports SAP connectors.

SAP Connectors

- SAP Table
- SAP ECC
- SAP HANA
- SAP BW

44. Real-Time Scenarios

Banking Scenario

- Extract transactions
- Validate records
- Load to data warehouse
- Generate reports

E-Commerce Scenario

- API ingestion
- Customer analytics
- Sales reporting

Healthcare Scenario

- Patient records integration
- Daily ETL processing

45. End-to-End Projects

Project 1: CSV to SQL Pipeline

Flow

1. Read CSV from Blob
2. Validate file
3. Copy data to SQL
4. Archive file
5. Send notification

Project 2: Incremental Load Pipeline

Flow

1. Read watermark value
2. Extract changed data
3. Load target table
4. Update watermark

Project 3: API to Data Lake

Flow

1. Call API
2. Store JSON
3. Convert to Parquet
4. Load analytics layer

46. Best Practices

Development Best Practices

- Use naming conventions
- Use parameterization
- Avoid hardcoding
- Implement logging
- Use reusable pipelines
- Use Git integration
- Enable monitoring

Performance Best Practices

- Use partitioning
- Use incremental load

- Optimize data flows
- Use staging carefully

47. Common Interview Questions

Beginner Level

1. What is Azure Data Factory?
2. What is Integration Runtime?
3. Difference between dataset and linked service?
4. What is a pipeline?
5. What are triggers?

Intermediate Level

1. Explain incremental loading.
2. Difference between ETL and ELT?
3. Explain mapping data flow.
4. How does self-hosted IR work?
5. Explain parameterization.

Advanced Level

1. How to optimize ADF performance?
2. Explain CI/CD in ADF.
3. How to handle pipeline failures?
4. Explain dynamic content.
5. How to secure ADF pipelines?

48. Developer Roadmap

Phase 1: Basics

- Azure basics
- Cloud concepts
- Storage concepts
- SQL fundamentals

Phase 2: ADF Fundamentals

- Pipelines
- Activities
- Datasets

- Linked services
- Triggers

Phase 3: Intermediate

- Data flow
- Dynamic content
- Incremental load
- API integration
- Error handling

Phase 4: Advanced

- CI/CD
- Databricks integration
- Synapse integration
- Security
- Performance tuning

Phase 5: Real Projects

- End-to-end projects
- Production scenarios
- Monitoring implementation

49. ADF Naming Standards

Recommended Naming Conventions

Component	Naming Example
Pipeline	PL_Copy_Customer
Dataset	DS_SQL_Employee
Linked Service	LS_SQL_Server
Trigger	TR_Daily_Load
Data Flow	DF_Transform_Sales

50. Important Terminologies

Term	Meaning
ETL	Extract Transform Load
ELT	Extract Load Transform
IR	Integration Runtime
CDC	Change Data Capture
DIU	Data Integration Unit
RBAC	Role-Based Access Control
CI/CD	Continuous Integration/Deployment

51. Common Errors and Fixes

Error: Authentication Failed

Fix

- Check credentials
- Check firewall access
- Verify linked service

Error: File Not Found

Fix

- Verify path
- Verify folder structure
- Check trigger timing

Error: IR Connection

Fix

- Restart IR service
- Check connectivity
- Verify gateway

52. Certification Guidance

Recommended Certifications

Microsoft Certifications

- DP-203: Data Engineering on Microsoft Azure

- AZ-900: Azure Fundamentals
- DP-900: Data Fundamentals

Preparation Strategy

1. Learn basics
2. Build projects
3. Practice labs
4. Solve interview questions
5. Attempt mock tests

53. Learning Resources

Official Resources

- Microsoft Learn
- Azure Documentation
- Azure YouTube Channel

Practice Platforms

- GitHub projects
- Hands-on labs
- SQL practice websites

Important Skills Along with ADF

- SQL 
- Python
- PySpark
- Azure Synapse
- Databricks
- Data Warehousing

54. Conclusion

Azure Data Factory is one of the most important cloud data integration tools for Data Engineers.

ADF helps organizations:

- Automate workflows
- Move large-scale data

- Build ETL/ELT pipelines
- Integrate multiple systems
- Create scalable cloud architectures

To become strong in ADF:

1. Learn fundamentals properly
2. Build hands-on projects
3. Practice real scenarios
4. Understand cloud concepts
5. Learn SQL deeply
6. Work on incremental loading
7. Understand monitoring and security

ADF combined with:

- SQL
- Python
- PySpark
- Synapse
- Databricks

creates a strong Data Engineering skill set.

Final Learning Sequence

1. Azure Basics
2. SQL Fundamentals
3. Azure Storage
4. Azure Data Factory Basics
5. Pipelines 
6. Activities
7. Triggers
8. Dynamic Content
9. Data Flow
10. Incremental Load
11. Error Handling
12. Monitoring
13. Git Integration
14. CI/CD
15. Synapse
16. Databricks
17. End-to-End Projects
18. Interview Preparation
19. Certification
20. Production-Level Architecture